

A Mid-air Multi-touch Interface using an Ultrafast Pan-tilt Camera

Yuta Abe
y.abe.796@ms.saitama-u.ac.jp
Saitama University
Saitama, Japan

Taishi Iriyama
iriya@mail.saitama-u.ac.jp
Saitama University
Saitama, Japan

Takashi Komuro
komuro@mail.saitama-u.ac.jp
Saitama University
Saitama, Japan

Kohei Shimasaki
simasaki@hiroshima-u.ac.jp
Graduate School of Advanced Science
and Engineering
Hiroshima University
Hiroshima, Japan

Idaku Ishii
iishii@robotics.hiroshima-u.ac.jp
Graduate School of Advanced Science
and Engineering
Hiroshima University
Hiroshima, Japan

ABSTRACT

We propose an interface that enables bare hand interaction with fingers of both hands even from a distance, using an ultrafast pan-tilt camera and a wide view camera. In this interface, a user can perform multi-touch interaction with a large display using mid-air finger motions. In order to acquire accurate finger motions required for multi-touch interaction, a wide view camera is used to detect the positions of the user's hands, and an ultrafast pan-tilt camera is utilized to obtain the images around the hands with high spatial resolution. In the experiments, we evaluated the accuracy of acquired finger motions, and showed that the proposed interface can capture both large motions of the user's whole body and small motions of the fingers with high accuracy. We also implemented two example applications, a drawing application and a photo layout application, using the proposed interface.

CCS CONCEPTS

• **Human-centered computing** → **User interface toolkits**; *Gestural input*; *Touch screens*.

KEYWORDS

active sensing, hand gesture recognition, bare hand interaction

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1 INTRODUCTION

In recent years, with the development of image processing technology, it has been possible to acquire the motion and posture of a person from a distance using RGB cameras[6, 8, 12] and depth sensors such as Kinect[3, 7, 14]. This enables user interfaces that

are operated by arm motions and hand gestures, and interactions such as object pointing[4, 16] and object manipulation[10] can be performed. However, these interfaces require large motions for interaction due to the limitations of the spatial resolution of the sensors. For detailed operations, such as selecting small objects or drawing precise figures, it is desirable for users to be able to perform interaction with small motions of fingers. LeapMotion is a device that can acquire accurate finger motions[9], and the interfaces using it enable detailed and multifunctional operations with fingers[15, 17]. However, it is difficult to acquire the accurate finger motions from a distance because high spatial resolution sensing is required.

In order to solve this problem, we propose an interface that enables bare hand interaction with fingers of both hands even from a distance, using an ultrafast pan-tilt camera and a wide view camera. In this interface, a user can perform multi-touch interaction with a large display using mid-air finger motions. In order to acquire accurate finger motions required for multi-touch interaction, a wide view camera is used to detect the positions of the user's hands, and an ultrafast pan-tilt camera is utilized to obtain the images around the hands with high spatial resolution. The ultrafast pan-tilt camera can simultaneously track multiple objects by capturing images at a high frame rate using a high-speed camera and high-speed gaze control using a pan-tilt mirror device.

2 RELATED WORK

2.1 Bare hand interaction

Bare hand interaction, which requires neither wearing the device nor holding the device in the hand, has advantages such as non-contact operation and eliminating the need for controller devices. To enable these interactions, methods using RGB cameras and depth sensors such as Kinect have been proposed. Liu et al. proposed a novel touchless visualization system for computer aided surgery, which enables a user to control and manipulate the patient's 3D anatomy model without contact through the use of Kinect-based gesture recognition technology[7]. Shen et al. proposed touchless interaction techniques for more natural and intuitive volume visualization using LeapMotion[15]. Nuzzi et al. proposed a real-time teleoperation system for robot end-effectors by acquiring the user's index finger position using RGB images and neural networks, and mapping the estimated position information to the robot's work

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space[12]. In these studies, these sensing systems can acquire accurate finger motions if the hands are within the detection range of the depth sensor or if they are captured with enough spatial resolution. However, it is difficult for a user to perform interaction from a distance because the resolution of the hand decreases with the distance between the hand and the sensing system.

2.2 High spatial resolution image acquisition by gaze control

In order to acquire detailed finger motions from a distance, the sensing system must be able to sense the hand with high spatial resolution. One method for acquiring high spatial resolution images is the use of a pan-tilt-zoom (PTZ) camera. A dual-camera system using a PTZ camera and a wide view camera has been used in some studies because it can track objects with a wide field of view and high spatial resolution. Neves et al. proposed a master-slave calibration algorithm capable of accurately estimating pan-tilt parameters without depending on additional constraints, and extend PTZ-assisted face recognition to surveillance scenarios[11]. Xu et al. proposed an autonomous surveillance system with multiple PTZ cameras assisted by a fixed wide view camera[18]. However, these methods require multiple PTZ cameras when tracking multiple targets, which may cause problems with camera control complexity and installation space depending on the number of cameras. In addition, in ordinary pan-tilt cameras, the weight and size of the camera head limits the actuator speed for pan-tilt movement.

In order to overcome this limitation, there is a study on a system using galvano mirrors to realize high-speed gaze control. This system enables high-speed pan-tilt movements by controlling the gaze of a high-speed camera using a mirror-based optical high-speed gaze control. Okumura et al. developed an optical high-speed gaze controller, called the “Saccade Mirror”, and used it to realize a ultrafast pan-tilt camera with a high-speed image processor[13]. Aoyama et al. proposed a method in which a single ultrafast pan-tilt camera with a galvano mirror functions as multiple virtual pan-tilt cameras[1]. By controlling each viewpoint with multi-threading and ultrafast viewpoint switching at the millisecond level, a single ultrafast pan-tilt camera enables simultaneous observation of different views. Hu et al. proposed a novel dual-camera system that can simultaneously capture zoomed-in images using an ultrafast pan-tilt camera and a fixed wide view camera using deep learning methods[5]. By detecting one object with the wide view camera and tracking it with each of the multiple virtual pan-tilt cameras, it is possible to simultaneously zoom in on multiple objects.

3 MID-AIR MULTI-TOUCH INTERFACE

3.1 Overview

An overview of the proposed interface is shown in Figure 1. The interface uses a large display that presents contents to the user, and a camera system consisting of an ultrafast pan-tilt camera and a wide view camera. When the camera system detects a user within its sensing range, the user is registered as the operating user and an operation window is placed in front of the operating user. The user can interact with the contents on the display through the operation window. The 2D position on the operation window corresponds to the position on the display. When a hand enters inside the operation

window, a hand cursor appears at the corresponding position on the display. The hand cursor can be controlled according to the motions of the user’s hand and fingers. The user can perform multi-touch operation by extending multiple fingers. When the user extends a finger, it is recognized as touching that position.

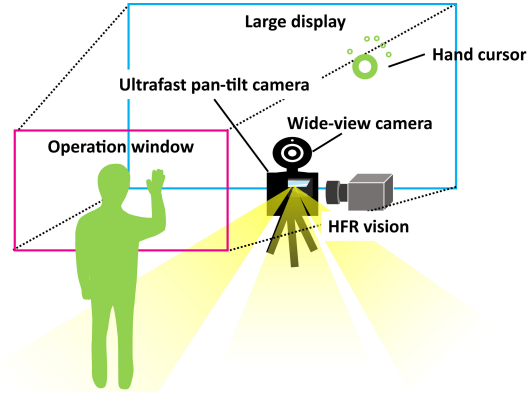


Figure 1: Mid-air multi-touch interface

3.2 Operation window placement and fixation in space

To enable users to interact from anywhere within the sensing range of the camera system in the same way, the operation window is placed in the proper position and of the proper size. The operation window is placed so that the center of the window is aligned with the user’s chest. The height of the window is calculated based on the distance from the user’s chest to waist, and the aspect ratio is set to the same ratio as that of the large display screen. The entire operation window is always within the user’s reach, allowing the user to interact with the entire large display without walking around. Each time the user’s posture is acquired, the center position and size of the operation window are updated. Therefore, the operation window may move due to the user’s movement during interaction. In this interface, the user’s raising a hand is set as the trigger that indicates the intention to perform precise interaction. When the system detects the triggering action, the operation window is fixed at that position. By keeping the operation window fixed during interaction, the user can interact with higher accuracy and stability than with a moving operation window. The fixation of the operation window is released when the user’s chest position is outside the window.

4 HIGH ACCURACY SKELETON DETECTION OF BOTH HANDS USING ULTRAFAST PAN-TILT CAMERA

In order to allow multi-touch interaction with the proposed interface, it is necessary to acquire accurate motions of fingers from a distance. For that purpose, a camera system using an ultrafast pan-tilt camera is used to perform high spatial resolution sensing of fingers of both hands. The captured images with high spatial resolution are used for high accuracy skeleton detection of both hands to acquire accurate finger motions.

4.1 Camera system

We use the camera system[5] shown in the Figure 2, which consists of a wide view camera for detecting the user and identifying the positions of both hands, and an ultrafast pan-tilt camera for gazing at the identified hand positions and acquiring telephoto images. The ultrafast pan-tilt camera has a camera lens with a longer focal length than that of the wide view camera. The ultrafast pan-tilt cameras can function as multiple virtual telephoto pan-tilt cameras by performing multi-threaded high-speed gaze control for each target.

The lens of the wide view camera has a focal length of 3.5 mm, and the lens of the ultrafast pan-tilt camera has a focal length of 25 mm. When capturing a scene 5 m in front of the camera, the wide view camera's 1440×1080 pixel wide view image is 5.6×4.2 m with a pixel size of 3.9 mm. The ultrafast pan-tilt camera's 480×640 pixel telephoto view image is 0.6×0.8 m with a pixel size of 1.25 mm. The magnification of the resolution of the telephoto view image relative to the wide view image is 7.1 times. The frame rate of the wide view camera is set at 60 fps, and the ultrafast pan-tilt camera is set at twice that rate, 120 fps.



Figure 2: Overview of the camera system

4.2 Hand skeleton detection of both hands using multiple virtual telephoto pan-tilt cameras

Figure 3 shows the method of hand skeleton detection of both hands using multiple virtual telephoto pan-tilt cameras. This method uses the wide view image to detect the user's whole body skeleton and acquire the positions of both hands. The positions of both hands on the XY image coordinate system of the wide view camera are converted to pan-tilt mirror angles. Based on the mirror angles, the user's hands are sensed with high spatial resolution by simultaneously gazing at them using multiple virtual telephoto pan-tilt cameras. Then, accurate skeleton detection of both hands is performed using the images of the telephoto views. OpenPose[2] is used as the skeleton detector for both the whole body and fingers.

Due to the processing time of the whole body skeleton detection using OpenPose and the blurring of the skeleton detection, the hand position acquired from the whole body skeleton detection and the hand position in real space at the time of gazing may be

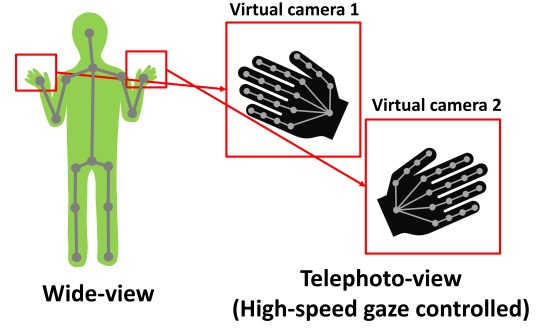


Figure 3: Hand skeleton detection of both hands using multiple virtual telephoto pan-tilt cameras

different. Therefore, it is necessary to synchronize them in order to use the large motions of the whole body and the small motions of the fingers together. The hand skeleton acquired from telephoto images is transformed into the coordinate system of the wide view camera using the hand positions that are used for gaze control. Then, the hand skeleton in the coordinate system of the wide view camera is integrated with the synchronized whole body skeleton. The hand skeleton detection compensates for the positional difference between the gazing position and the real hand position, allowing to capture both large motions and small motions.

5 ACCURACY EVALUATION

We conducted an experiment to verify that the system can acquire accurate finger motions of both hands from a distance and to perform multi-touch interaction. In the experiment, a user interacted using both hands at a distance of 5 m away from the camera system and the large display, as shown in Figure 4.

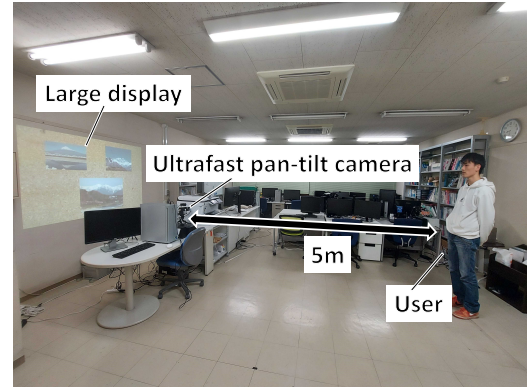


Figure 4: Experiment environment

We verified if the proposed method can acquire accurate finger motions of both hands and detect which fingers are extended, even from a distance. In the experiment, we compared the results of hand skeleton detection for 12 different hand postures by one subject and the detection of the extended fingers using the proposed method and a method that uses only a wide view camera. The experiment

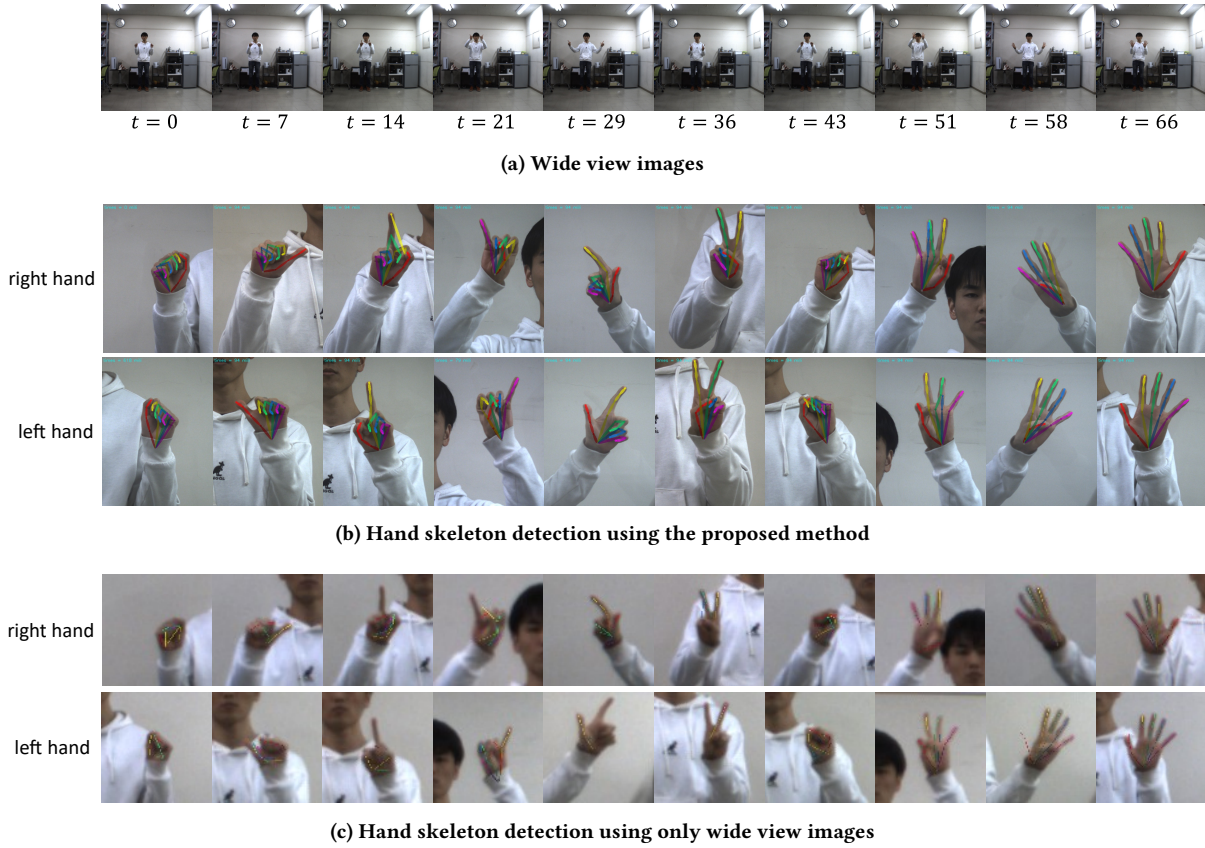


Figure 5: Comparison of accuracy of hand skeleton detection between the proposed method and using only wide view images

lasted about 67 seconds, and a total of 826 frames of the left and right hands were captured. In creating the ground truth labels for extended fingers detection, most frames were manually labeled by one of the authors. In the frames around hand posture change, where it is more difficult to determine the extended fingers, several colleagues from the same laboratory checked the labels for errors. The fingers were detected as extended if the fingertips were more than a certain distance away from the skeletal point at the base of the middle finger.

Figure 5 shows the wide view camera images every 7 seconds during the experiment and the results of hand skeleton detection using the proposed method and the method using only wide view image at the corresponding time. When only the wide view camera was used, the hand skeleton detection failed in most cases, except for the postures in which all fingers were extended and the posture in which only the index finger was extended. In the case of using the proposed method, the hand skeleton detection was successful in most of the postures. On the other hand, in the posture with only the middle finger extended and in the posture with the middle, ring, and pinky fingers extended, the hand skeleton detection was failed. In these postures, the middle finger was recognized as the index finger, and there was room for improvement in the hand skeleton detection method. When the middle and index fingers

were extended at the same time, these recognition failures were less likely to occur, and detection was successful.

Figure 6 shows the detection rate of extended fingers for each posture using the same images as input. When only a wide view camera was used, it was not possible to detect extended fingers in almost all postures, and the overall detection rate was 25.3%. The proposed method correctly detected extended fingers with an overall detection rate of about 77.8%. Excluding the postures in which only the middle finger was extended and the posture in which the middle, ring, and pinky fingers were extended, the proposed method was able to detect the extended fingers in about 89.4% of the cases. In most cases where skeleton detection of both hands was successful, extended fingers were detected correctly.

We also conducted an experiment to evaluate the effectiveness of synchronizing the skeleton detection results of the whole body and hands. In this experiment, we compared the trajectories of the fingertips with and without synchronization by moving the left hand in a figure-eight motion with the hand open. The differences in the trajectories of the fingertips with and without synchronization are shown in Figure 7. In both cases, the relative positions of the fingertips and the rough overall movements were the same. In the case without synchronization, the hand position and hand posture did not match well, resulting in a significant disruption of the trajectory. In the case with synchronization, high accuracy hand

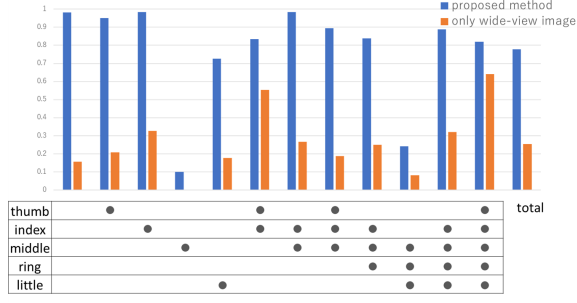


Figure 6: Detection rate of extended fingers for each posture

skeleton detection compensated the position, and the trajectory matched the actual finger motions.

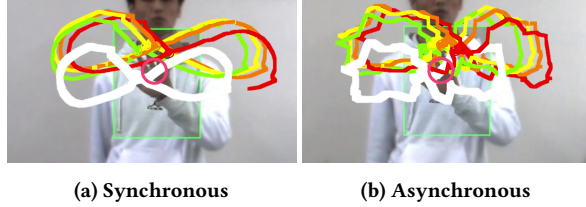


Figure 7: Comparison of hand position accuracy between synchronous and asynchronous cases

6 APPLICATION EXAMPLES

The proposed interface enables multi-touch interaction from a distance using both large motions of the whole body and small motions of the fingers. We implemented a system based on the proposed interface. Figure 8 show the hand cursor superimposed on a cropped image of the area in the operation window. A hand cursor consists of points indicating each fingertip and a large circle indicating the center of the hand. The center of the hand is defined as the skeletal point at the base of the middle finger. The finger is detected to be extended if the distance from the center of the hand to the fingertip is greater than a certain distance.

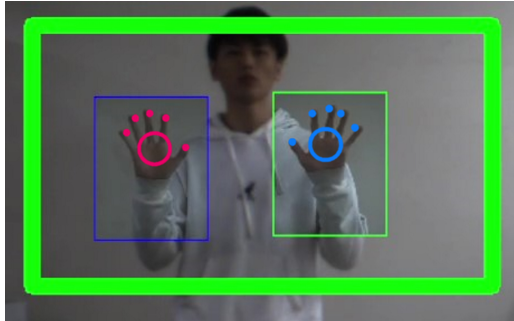


Figure 8: The hand cursors during interaction

We show the multi-touch interaction with two example applications using the proposed interface. We implemented two example

applications: a drawing application and a photo layout application. In the drawing application, each finger (except the thumb) is assigned a different color, and by touching the screen, the user can freely draw lines with each finger. The thumb is assigned as eraser, and drawn lines can be erased by dragging over the lines with the thumb. In addition, the drawing application allows users to draw multiple lines simultaneously and to draw precise figures using different line colors. In the photo layout application, the application starts with several photos initially placed, and by dragging the photo with one finger, the photo can be moved by following the fingertip. By using more than one finger, it is possible for the user to zoom in and out and rotate the touched photo. The photo layout application can be interacted easily with only one hand by using both the large motions of the arm and the small motions of the fingers.

Figure 9 shows the interaction of two applications using the proposed interface. Figure 9a and 9b show the interactions using the drawing application. The user used multiple fingers to draw multiple lines simultaneously, and to draw precise figures using different line colors. The smoothness of the lines showed that the application was capable of precise interaction. In addition, interactions using small motions of fingers allowed users to perform detailed operation, such as drawing precise figures. Multi-touch interactions such as simultaneously drawing lines and quickly switching colors were possible. Figure 9c shows moving of photos by single touch using the photo layout application. The user could interact with all the photos on the screen without walking around. Figure 9d shows zooming in and out and rotation of photos by multi-touch interaction. The same interaction could be performed with only one hand in less motion than with both hands. Especially for rotation, the arm-twisting motion made it possible to adjust the angle more intuitively than using both hands.

7 CONCLUSION

We proposed an interface that enables bare hand interaction with fingers of both hands even from a distance, using an ultrafast pan-tilt camera and a wide view camera. In this interface, a user can perform multi-touch interaction with a large display using mid-air finger motions. In order to acquire accurate finger motions required for multi-touch interaction, a wide view camera is used to detect the positions of the user's hands, and an ultrafast pan-tilt camera is utilized to obtain the images around the hands with high spatial resolution. In the experiments, we evaluated the accuracy of acquired finger motions, and showed that the proposed interface can capture both large motions of the user's whole body and small motions of the fingers with high accuracy. We also implemented two example applications, a drawing application and a photo layout application, using the proposed interface.

There are three future works. The first is the extending the system to the multi-user operable interface. Hand skeleton detection within this interface increases processing time depending on the number of hands. Therefore, it is necessary to perform processing optimization and system optimization based on the number of operating users. The second is realizing new user experiences in the multi-user operable interface. By utilizing the fact that other users' operations affect their own operations, it is expected that

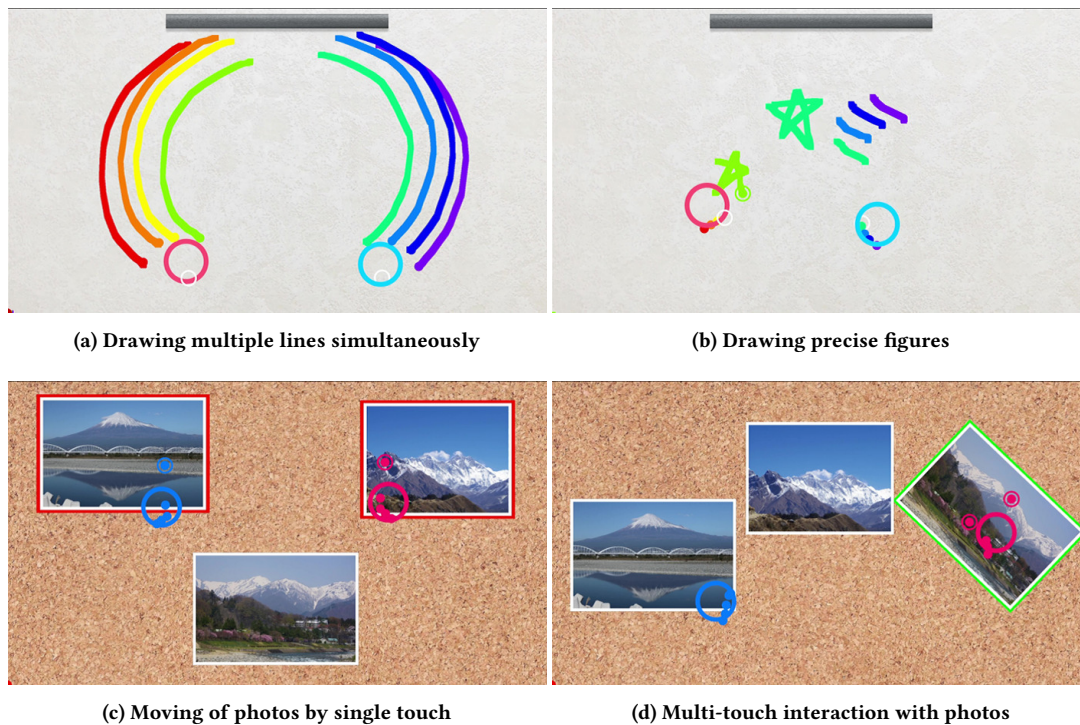


Figure 9: Drawing and photo layout applications

this interface can be used for effective information presentation in digital signage. The third is an evaluation of the usability of this interface. By implementing an interaction technique triggered by the extended finger, this interface enabled users to perform accurate operations from a distance. In the future, it is necessary to conduct a comparative evaluation of usability and fatigue with existing interfaces through user study.

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